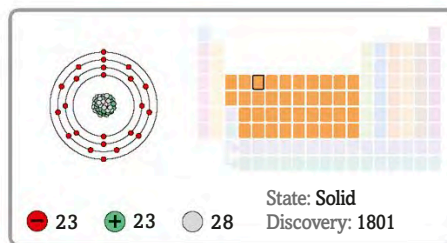


23

V

Vanadium



Forms



This mushroom contains high levels of vanadium.

Fly agaric mushroom



These brittle crystals are the main source of vanadium.



This powdery yellow crust contains tiny amounts of vanadium.

Silvery surface

Laboratory sample of pure vanadium crystals



Uses



Tools made with alloys of vanadium and steel are durable.

About
85% of all vanadium
is used to
toughen steel.

This knife has been strengthened by the addition of vanadium.



Vanadium can be hammered and stretched without breaking. This hard, strong metal is easy to shape. **Vanadium** was first **purified** in 1869 by the British chemist Henry Roscoe. Today, it is commonly extracted from its ore **vanadinite**. Ancient metalworkers used tiny

amounts of vanadium compounds to make a very tough substance called **Damascus Steel**. This was named after the capital city of Syria, where ironworkers made the world's sharpest swords. Vanadium is still used to toughen tools, such as **spanners** and knives.